D Y PATIL DEEMED TO BE UNIVERSITY — RAMRAO ADIK — INSTITUTE OF TECHNOLOGY NAVI MUMBAI

D Y Patil University

Ramrao Adik Institute of Technology

LAB MANUAL

Engineering Physics

List of Experiments

Sr. No.	Name of Experiment
1	Newton's Rings :Determination of radius of curvature of a Plano- convex lens
2	Divergence of Laser beam and Determination of wavelength of laser beam
3	Optical fibre : Determination of Numerical aperture
4	Determination of Planck's constant 'h' using LED.
5	Energy Band Gap of Semiconductor
6	Hall Effect: Determination of Hall coefficient and Carrier Concentration.
7	Ultrasonic Interferometer
8	Study of Scanning Electron Microscope (SEM) and Atomic Force Microscopy (AFM)
9	Virtual Lab
10	Virtual Lab

Name of the Student:	
Division:	
Roll No.:	

Experiment 1

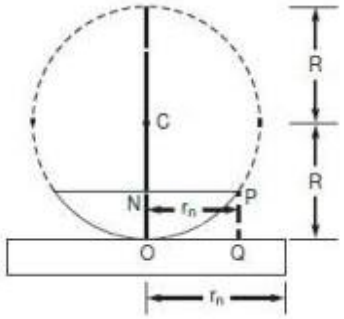
1. **Aim:** To measure the radius of curvature of Plano convex lens using Newton's rings set up.

CO1	Study of Newton's rings by the technique of division of amplitude in thin film interference.

2. **Apparatus:** Travelling microscope, Plano convex lens of large radius of curvature, glass plates, Lamp on stand, sodium vapour lamp, and magnifying lens.

3. **Theory:**

A Plano convex lens of large radius of curvature is placed on a clean glass plate (optically flat) and viewed in monochromatic light, a series of rings may be seen around the point of contact between the lens and the slide. These rings are known as Newton's rings and they arise due to the interference of light reflected from the upper surface of glass plate of the air film and lower surface of lens. Under reflected system the interference pattern is alternate dark and bright concentric circles, with centre dark spot.



Suppose the radius of curvature of the lens is R and the thickness of the air film is t at distance OQ = r, from the point of contact O. The effective path difference between the interfering rays is

$$\Delta = 2\mu t \cos r \pm \frac{\lambda}{2}$$

The condition for constructive and destructive interferences are given as; for normal incidence **cos r = 1** and for air film **μ = 1.**

For constructive interference $2t = (2m+1)\frac{\lambda}{2}$

For destructive interference $2t = m\lambda$

The thickness of the thin film in terms of any dark or bright ring is given by

$$2t = \frac{r_m^2}{R}$$

Central dark spot: At the point of contact of the lens with the glass plate the thickness of the air film is very small compared to the wavelength of light therefore the path difference introduced between the interfering waves is zero. Consequently, the interfering waves at the centre are opposite in phase and interfere destructively. Thus a dark spot is produced.

Circular fringes with equal thickness: Each maximum or minimum is a locus of constant film thickness. Since the locus of points having the same thickness fall on a circle having its centre at the point of contact, the fringes are circular.

1. Radii of the mth dark rings: $r_m = \sqrt{m\lambda R}$

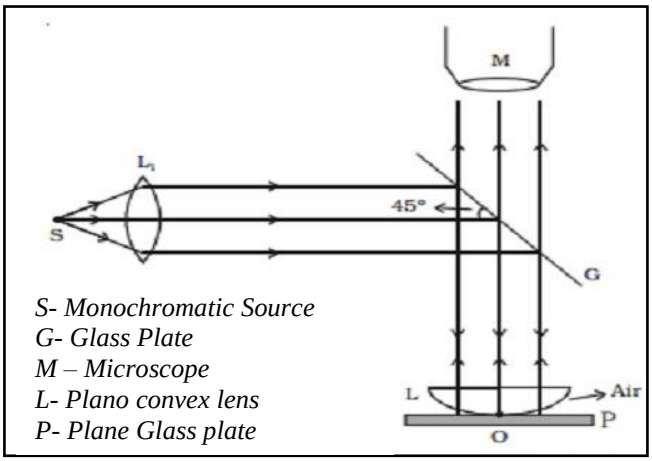
2. Radii of the mth bright ring: $r_m = \sqrt{(2m+1)R\frac{\lambda}{2}}$

The wavelength of monochromatic light can be determined as,

$$\lambda = \frac{D_{m+n}^2 - D_n^2}{4mR}$$

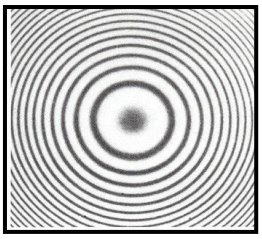
Where, D_{m+n} is the diameter of the (m+n)th dark ring and D_n is the diameter of the nth dark ring.

Diagrams:



S- Monochromatic Source
G- Glass Plate
M – Microscope
L- Plano convex lens
P- Plane Glass plate

Experimental Set up



Reflected System

4. **Procedure:**
   1. Illuminate the Newton's ring set up with monochromatic Sodium light.
   2. View the light reflected from air film through eyepiece of the travelling microscope. Observe the interference pattern which contains concentric circular dark and bright rings with centre dark spot.
   3. Bring cross wire towards left 10th dark ring, note the reading.
   4. Shift the cross wire towards 10th, 8th, 6th, 4th , 2nd on L.H.S. Similarly move towards 2nd, 4th, 6th, 8th and 10th on R. H. S. Accordingly note the readings.
   5. The difference between LHS and RHS readings give the diameter of particular order of any dark ring.

Observations:

Least Count of Travelling Microscope = ...mm

Wavelength of Sodium light = λ = 5893 Å = 5893×10^{-7} mm

Sr. No.	Dark ring no.	Microscope reading (mm)		Diameter D (mm)	D^2 (mm^2)	$(D_{m+n})^2 - (D_n)^2$ (mm^2)
		L.H.S.	R.H.S.			
1.	10					
2.	8					

3.	6					
4.	4					
5.	2					
						Avg =

Formula: Radius of curvature,

R by calculation = $\frac{(D_{m+n})^2-(D_n)^2}{4m\lambda}$ **m=2**

R=

6. Result

1. Radius of curvature of Plano convex lens, R =..... mm =.....m by calculation

7. Conclusion and Discussion:

Central Spot is dark in reflected light due to phase inversion of reflected ray (Stoke's law) and Rings come closer and become thinner with increasing order.

8. QUIZ / Viva Questions:

1) What change in the pattern will be seen if monochromatic light is replaced by white light?
2) How one can get bright spot at the centre of pattern in reflected system?
3) What happens if air film is replaced by water?

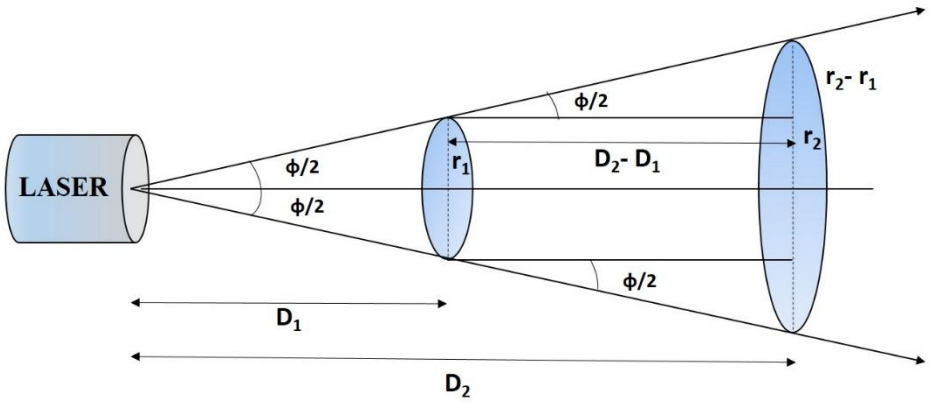
Experiment 2

1. **Aim:** i) To measure the peak power and beam divergence of a given laser beam
   ii) To determine wavelength of LASER using diffraction grating.

CO2	Study of characteristics and application of laser source with help of diffraction

2. **Apparatus:** He-Ne laser, Optical bench, Laser Beam Analyzer (LBA) with sensor and micrometer screw arrangement.

3. **Diagram:**



Divergence of laser beam

4. **Formula:**

Divergence of laser beam is given by,

$$\emptyset = \frac{D_2 - D_1}{d_2 - d_1}$$

Where, **'D'** is diameter of laser spot for the distance 'd' between LBA and Laser.

5. **Theory**

Laser is an extremely coherent, monochromatic, directional, focusable, polarized and powerful light. These extraordinary features make it greatly applicable in day-to-day life, science and technology. A few notable applications of laser include

medical diagnosis and treatments, fiber optic communications, CD-ROMS, CD players, laser printers, defense, cutting, welding. drilling, surveying, aligning etc.
Laser is produced due to stimulated radiation; a process where a resonating photon stimulates the de-excitation of an excited atom. This results in to emission of two coherent photons, which are identical in all respects. These photons further stimulate the de-excitation of other excited atoms and this continues to generate an avalanche of coherent photons. For stimulated emission to take over spontaneous emission and stimulated absorption, a conditions are necessary. These are availability of metastable state (life time = 10 sec). population inversion (greater number of atoms in metastable state than in lower energy state) and enough number of photons in the cavity (mirrors). a few

He-Ne laser

He-Ne laser is a low power, continuous gas laser, which is used in supermarket scanners, student laboratories and holography. The active system is neon, which is pumped electronically via helium in a resonant cavity made of discharge tube (Fig. 5.1). The main lasing occurs in neon between the levels E6 (metastable) and E3 which produces an intense coherent beam of red color (wavelength 6328Å). (refer Fig 5.2). The population of photons necessary for stimulated emission is maintained by mirrors (one is semitransparent) on both sides. Brewseter windows are used to polarize the laser light.

6. Procedure:

1. Make the laser beam ON. Avoid eye contact
2. The power in laser beam follows Gaussian distribution with peak value at center.
3. Mount the sensor of LBA on optical bench at a distance relatively closer to laser beam, say 10 cm. Let this distance be d_1. Adjust sensor so that laser is incident exactly on the centre of the window of the sensor. Align the sensor, till LBA reads power closest to 2.0 mW.
4. Now move the sensor laterally so that the beam falls on the edge of the window of the sensor. LBA will now read zero.
5. Using micrometer screw, move the sensor-window gradually across the laser beam. Note the increasing powers in the beam (mW) at various screw positions (mm) as per table 5.1. At certain stage, the power in LBA will reach peak and then will start decreasing, even though the screw is moved in the same direction. Note the

decreasing powers at various advanced screw positions.
Note that the screw should be moved in only one direction throughout the observations.

6. For measuring the screw positions. use following procedure
   Screw position = X = MSR + VSR x LC mm
   Where MSR is the reading on the main scale, which is closest to the edge of the screw. VSR is the vernier scale reading, which is the sequence number of the division on the screw which coincides with the line on main scale.

   LC is the least count of micrometer screw gauge
   LC = smallest division on the main scale / number of divisions on the vernier sale = 0.5 mm / 50 = 0.01 mm
7. Repeat the entire procedure from 2 to 4, by placing the sensor at d_2 cm, sufficiently away from d, (say by 50 cm). Record these observations in table 2
8. Plot the graph of power (mW) Vs position (mm) for observation table 1 (for d_1). Identify the peak power (Pm). Also identify a point on power axis corresponding to Pm/2. Draw a horizontal line starting from Pm/2. This line will intersect the Gaussian curve at two points having X co-ordinates X1 (mm) and X2 (mm). The quantity D (mm) = (X2-X1) i.e. Full Width at Half Maximum (FWHM) gives the effective diameter of laser when the distance between LBA and laser is d1 cm. (refer sample graph)
9. Plot the graph of power (mW) Vs position (mm) for observation table 2 (for d2). Repeat the procedure explained in step 6 and calculate the diameter D2 (mm) of the laser beam at the position d2. (refer a widened graph)
10. The Gaussian distribution at the position d_2 will be slightly wider than that at position di. Consequently the diameter D_2 of the laser beam at the position de will be greater than diameter D at the distance d_1. Calculate the divergence of laser beam by using the formula and procedure in 'Calculations'

7. Observation Table

Table 1 Powers at different positions at a distance D_1= ….cm			Table 2 Powers at different positions at a distance D_2= …..cm		
Sr. No.	**Power in LBA P (mW)**	**Position of micrometer X (mm)**	**Sr. No.**	**Power in LBA P (mW)**	**Position of micrometer X (mm)**
1	0		1	0	
2	0.25		2	0.25	
3	0.5		3	0.5	
4	0.75		4	0.75	
5	1		5	1	
6	1.25		6	1.25	
7	1.5		7	1.5	
8	1.75		8	1.75	
9	Peak=		9	Peak=	
10	1.75		10	1.75	
11	1.5		11	1.5	
12	1.25		12	1.25	
13	1		13	1	
14	0.75		14	0.75	
15	0.50		15	0.50	
16	0.25		16	0.25	

8. Graphs

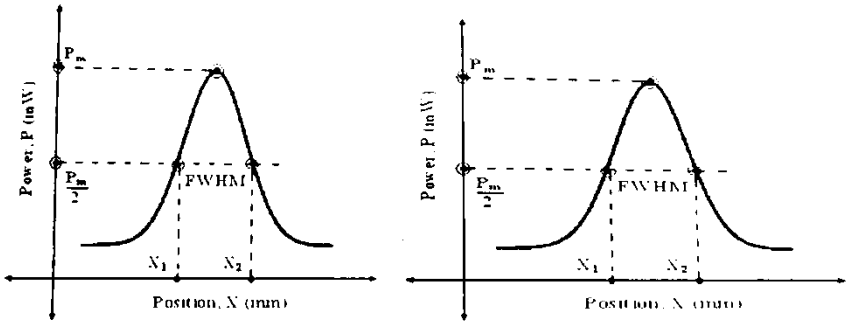


Figure 5.3 Calculation of (a) diameter *D1* at position *d1 (b)* diameter *D2* at position *d2*

9. Calculations:

$$\emptyset = \frac{D_2 - D_1}{d_2 - d_1}$$

10. Result and Conclusion:

1. Divergence of LASER Beam:

2. Wavelength of laser Beam:

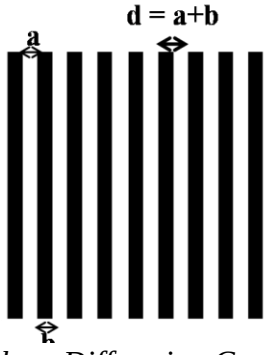
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ii) Determination of wavelength of Laser (Demo experiment should not written in journal)

1. **Aim**: To determine wavelength of LASER using diffraction grating.
2. **Apparatus:** He-Ne gas Laser, Power supply, Plane diffraction grating, and Scale.
3. **Theory**

Diffraction Grating:

An arrangement consisting of a larger number of parallel slits of the same width, separated by equal opaque spaces is known as DIFFRACTION GRATING. When a Wave front is incident on a grating surface; light is transmitted through these slits but obstructed by opaque spaces. Such a grating is called transmission grating. There is another type of grating called as refection grating.



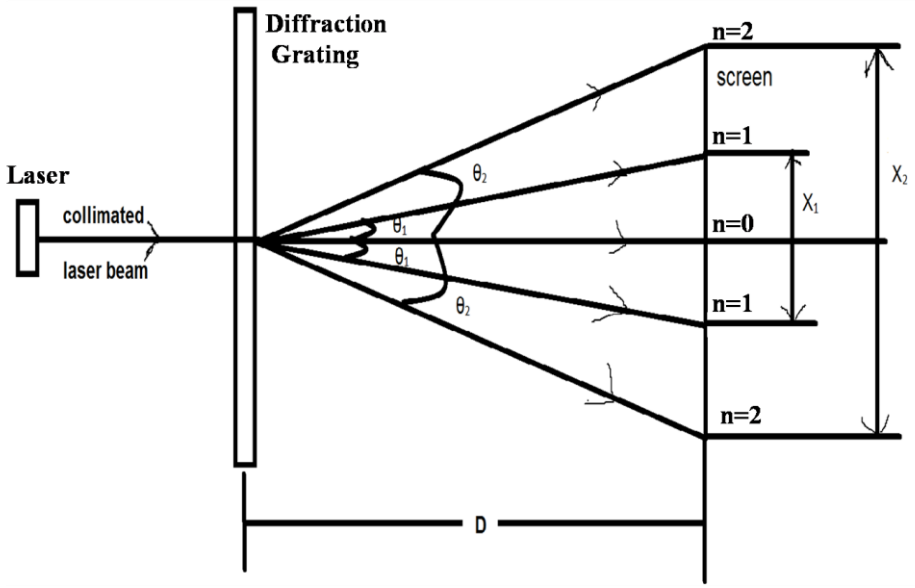
Plane Diffraction Grating

4. **Formula**

(a + b) sin θ = m λ,

λ= (a + b) sin θ / m. where a+b = d (grating element)

5. **Diagram:**



6. **Observation**: The number of lines per mm on the grating, N = 300 lines / mm

Hence, grating element, d= a+b = 1 / N =...1/300.....mm

7. **Procedure:**

1) Set up the laser and grating as shown in figure.
2) Measure the distance D between the Screen and the diffraction grating.
3) Turn on the laser and center the meter stick at the 0th order image on the screen.
4) Measure the distance X1 between the 1st order images appearing on the left and right sides of the center line.
5) Measure the distance X2 between the 2nd order images appearing on the left and right sides of the center line.

8. Observation Table:

(Write down values of diffraction angle up to two decimal points and sin, tan up to four decimal points.)

Sr. No.	Distance between the grating element and screen' D' (cm)	Order of diffraction 'm'	Distance between m^{th} order spots 'X'(cm)	$\tan\theta = X/2D$	θ (deg)	$\sin\theta$	$\lambda = \frac{d\sin\theta}{m}$ (Å)
1							
2							
3							
						Mean λ =	

9. Calculation:

Wavelength of Laser source is given by,

$$\lambda = \frac{d\sin\theta}{m}$$

10. Result: Average Wavelength of laser beam, λ = ...mm =---------- Å.

11. Conclusion and Discussion:

The given source of light is highly monochromatic and coherent in nature.

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12. QUIZ / Viva Questions:

1. Define following terms
   Stimulated emission, Spontaneous emission, Stimulated/resonance absorption, metastable state, Population inversion, Pumping, Active system, Resonant cavity, Brewster window
2. What is the role of He in the action of He-Ne laser?
3. What is the role of Ne in the action of He-Ne laser?
4. Is He-Ne laser a continues laser or pulsed laser?
5. Ideally laser is supposed to move as a parallel beam. Why does it diverge then!
6. Which element is responsible for red light of He-Ne laser? He? Or Ne?
7. Does He-Ne laser emit only 6328 A°. Or other wavelengths also? If, yes then what are these wavelengths?
8. What are the advantages of He-Ne laser!
9. What are the disadvantages of He-Ne laser?
10. What are the applications of He-Ne laser?
11. Why He atoms are at quite a higher percentage and quite a high pressure than Ne atoms?
12. Why does He-Ne laser require heavy and high tension power supply?Viva
13. Why does He-Ne laser require heavy and high tension power supply?What do you mean by diffraction? Why don't we observe diffraction of sound in daily life?
14. How Young's double slit interference differ from double slit diffraction in terms of dimensions and wavelength?
15. How separation between the spectral lines will be increased in diffraction pattern obtained in white light using diffraction grating?

Experiment 3

1. **Aim:** To determine the numerical aperture, acceptance angle, V number and number of modes of the given optical fibre
2. **Apparatus:** Fibre optic kit, N.A. Measurement jig, 1m optical fibre.
3. **Theory:**

   1) An optical fibre (or optical fibre) is a flexible, transparent fibre made by drawing glass (silica) or plastic to a diameter slightly thicker than that of a human hair.

   2) Optical fibres typically include a transparent core surrounded by a transparent cladding material with lower refractive index (R. I.). Light is kept in the core by the phenomenon of total internal reflection which causes the fibre to act as a waveguide.

   3) Optical fibres are used most often as a means to transmit light between the two ends of the fibre and find wide usage in fibre-optic communications, where they permit transmission over longer distances and at higher bandwidths (data rates) than wire cables

   Numerical aperture (N. A.) parameter provides maximum light handling capacity of the optical fibre. It refers to the maximum angle (acceptance angle) at which the light incident on the fibre end is totally internally reflected and is transmitted properly along the fibre.

   This is given by:

$$N.A. = Sin(i_{max})$$

4. **Formula:**

   Numerical aperture:

$$N.A. = Sin(i_{max})$$

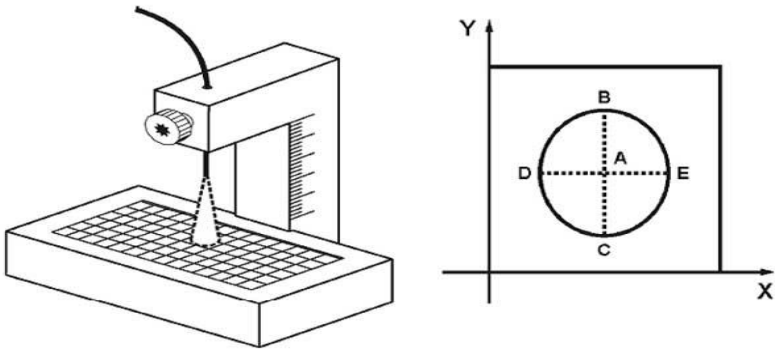
   V Number:

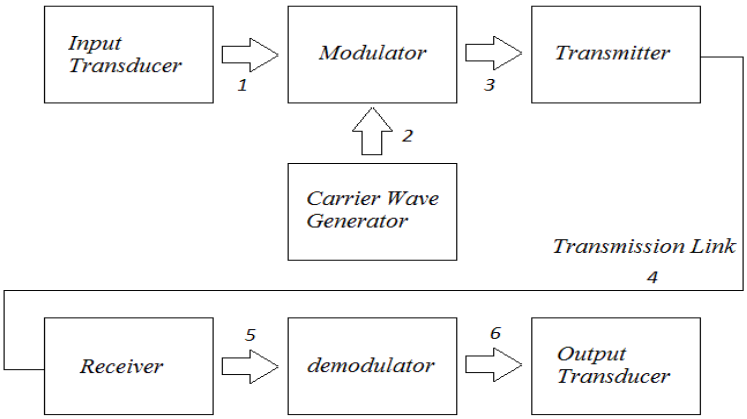
$$V = \frac{\pi d}{\lambda}\ N.A.$$

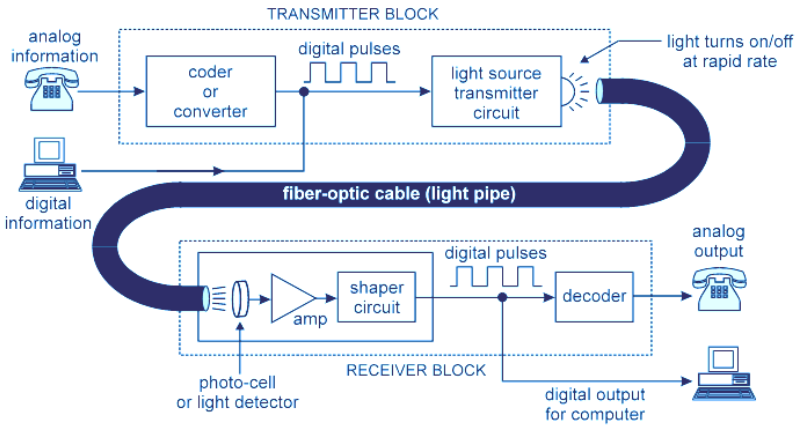
   No. of modes for Step Index Fiber:

$$N_m = \frac{V^2}{2}$$

5. Diagrams:







Optical Fiber communication system

6. Procedure: :

1) Connect the optical fibre to kit.
2) Connect the power supply to kit and switch it 'On', observe a glow at the other end of the fibre.
3) Insert the other end of the optical fibre into the N.A. measurement jig.
4) Light should fall normally on the white sheet.
5) Observe the illuminated circular path of light on the sheet.
6) Measure the distance between the sheet and the fibre exactly ('a' cm).
7) Calculate the mean radius ('r' cm) of the illuminated patch.
8) Determine the value of Numerical Aperture (N.A.) and Acceptance angle (i_{max}).

Observations:

1. Wavelength of the Laser:

2. Diameter of the core = d =

Observation Table:

Sr. No.	Distance between Fiber tip and Screen "a" (cm)	Radius of the Circular Spot "r" (cm)	Acceptance Angle i_{max} (degree)	Numerical Aperture (N. A.)
1.				
2.				
3.				
4.				
5.				

6. Calculations:-

$$\tan(i_{max}) = \frac{r}{a}$$

$$i_{max} = tan^{-1}\left(\frac{r}{a}\right)$$

$$N.A. = Sin(i_{max})$$

7. Results:

1. Maximum angle of incidence, i_{max} = ---------- degree

2. Average N.A. of the given fiber is = ----------

8. Conclusion and Discussion:

Numerical aperture of given fibre is greater than 0.2, therefore fibre is MMF**.**

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9. QUIZ / Viva Questions:

1) What do you mean by total internal reflection? How it works in Optical fibres?
2) How acceptance angle and critical angle are related?
3) Explain basic optical fibre communication system?
4) What are the demerits of optical fibre communication system?

Experiment 4

1. **Aim:** Determination of Planck's constant using Light Emitting Diode (LED).

2. **Apparatus:** Different colours LEDs, Power supply, mili Voltmeter, mili Ammeter and connecting wires.

3. **Theory:**

Planck's constant:

The Radiation energy emitted or absorbed is discontinuous (discrete) and it is in the form of quantum (energy packet). Planck's constant is the fundamental constant in modern physics. It relates the energy of a photon to its frequency.

$$E = h\gamma$$

Where, E is the energy of photon, h is the Planck's constant and γ is the frequency of the emitted photons. When the diode first emits light the voltage across the diode, V_0, is just enough to give energy to electrons to jump from valance band to conduction band. Therefore

$$E = eV_0 = h\gamma = \frac{hc}{\lambda}$$

Where e is the electron charge and V_0 is the threshold voltage.

LED (Light Emitting Diode):

A light-emitting diode (LED) is a p–n junction diode (made up form direct band gap semiconductor such as GaAs, GaP and GaN) emits light in forward bias. When a suitable voltage is applied to junction, electrons are able to recombine with holes within the device, releasing energy in the form of photons. The output from an LED can range from red (at a wavelength of approximately 700 nanometers) to blue-violet (about 400 nanometers)

Conventional LEDs are made from a variety of inorganic semiconductor materials, producing the following colours:

1. Aluminium gallium arsenide (AlGaAs) — red and infrared
2. Aluminium gallium phosphide (AlGaP) — green
3. Gallium arsenide phosphide (GaAsP) — red, orange-red, orange, and yellow
4. Gallium phosphide(GaP) — red, yellow and green

4. Procedure:

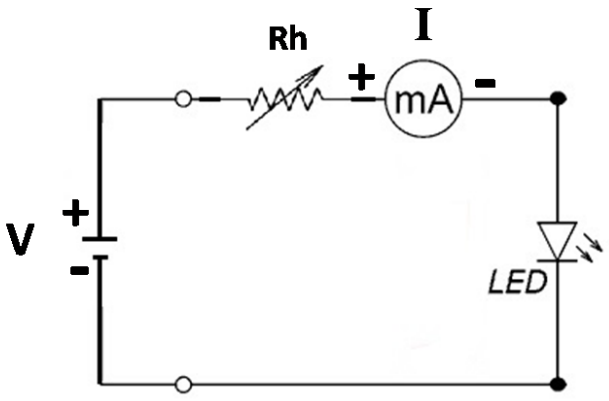
1. Make the connection on experimental kit as shown in circuit diagram.
2. Set the range of DC voltmeter at 20 mV and switch 'On' the power supply.
3. Now vary the DC voltage slowly by variable resistance pot and note current readings.
4. Now switch 'Off' the DC power supply and break the LED connection.
5. Again make same connection for another colour of LED.
6. Now repeat the above steps for different LEDS.
7. Repeat above experiment for different colours of LEDs.
8. Plot I-V characteristics for different LEDs and find out threshold voltage $V_{0.}$
9. Now use the formula given below and put the value of all parameters used in formula and calculate the value of Planck's constant for different LEDs.

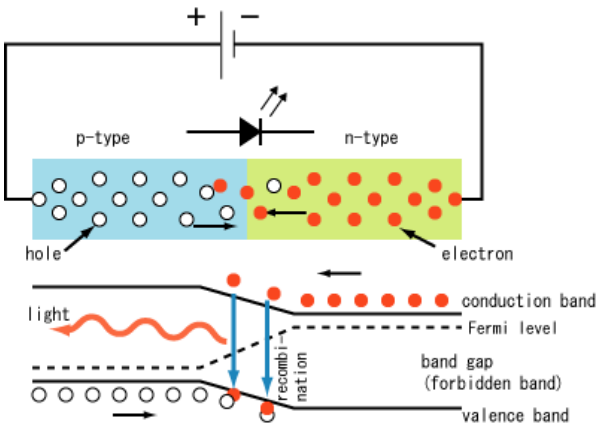
$$E= h f=eVo=hc/ \lambda$$

$$h = \frac{eV_0\lambda}{c}$$

10. Take mean value of h calculated for different LEDs.

5. Diagrams:





6. Formula:

$$h = \frac{eV_0\lambda}{c}$$

where, h - Plank's constant

e - Charge (1.6×10^{-19} C)

V_0 - Threshold Voltage

λ - Wavelength of radiation

c- velocity of radiation (3×10^8 m/s)

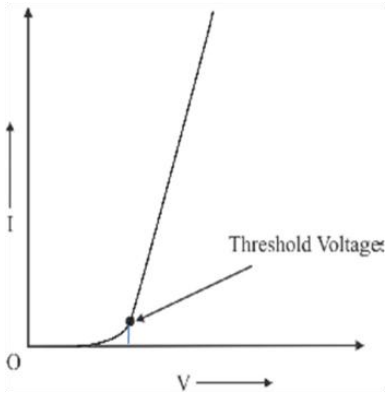
7. Observation Table:

1. For Red LED (λ = 700 nm) 2. For Green LED (λ= 560 nm) 3. For Yellow LED (λ= 580 nm)

Sr. No.	Voltage **(V)**	Current **(mA)**	Sr. No.	Voltage **(V)**	Current **(mA)**	Sr. No.	Voltage **(V)**	Current **(mA)**
1.			1.			1.		
2.			2.			2.		
3.			3.			3.		
4.			4.			4.		
5.			5.			5.		
6.			6.			6.		
7.			7.			7.		
8.			8.			8.		
9.			9.			9.		
10.			10.			10.		
11.			11.			11.		
12.			12.			12.		

8. Graph:

Plot graph of Voltage **V** (X axis) vs. current **I** (Y axis) for different LEDs.



Find out value of Threshold voltage for each LED from the graph.

9. Calculations:

1. $h = \frac{eV_0\lambda}{c}$

11. Mean h =

10. Conclusion:

1. Planks constant h = Js (Experimental)
2. Planks constant h = 6.626 x 10^{-34} Js (Theoretical)

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Viva Questions:

1) What do you mean by quanta and Plank's constant?
2) Why Si or Ge semiconductors are not used in LEDs?
3) Why LED gives monochromatic light and how do we get different colours in LED?

Experiment 5

1. **Aim:** To determine the energy gap of a germanium semiconductor diode.

2. **Apparatus:** Energy gap trainer kit, micro ammeter, digital multimeter, water heater, connectors.

3. **Theory:**

One of the simplest methods of determining the energy gap E_g of a semiconductor material involves the study of variation in reverse saturation current of a p n-junction diode. The reverse saturation current is represented mathematically by the expression

$$\mathbf{I_s = AT^2 exp\ (-E_g / k\ T)}$$

Where, I_s= reverse saturation current, A=material constant, E_g= energy gap, K= Boltzmann's constant &T= Absolute temperature of the junction.
Thus,

$$\mathbf{I_s / T^2 = A\ exp\ (-E_g / k\ T)}$$

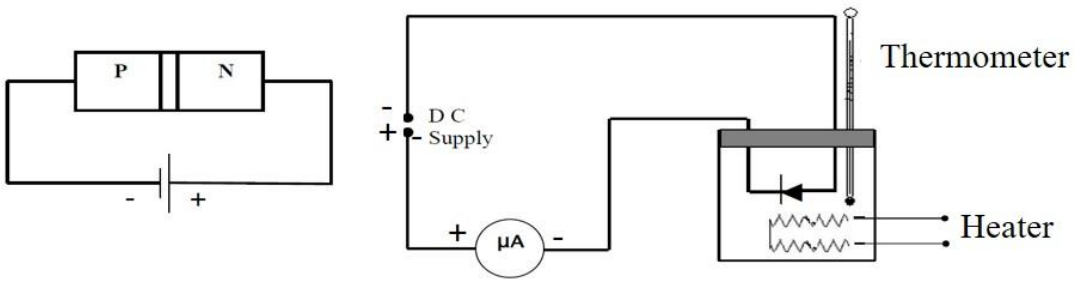
$$\mathbf{\ln (I_S/T^2) = (-E_g / k\ T) + \ln A}$$

And,

$$\mathbf{\log_{10} (I_S/T^2) = (-E_g / 2.303\ k\ T) + constant}$$

Hence from the graph of $\log (I_S/T^2)$ Vs. (-1/T) the energy gap can be calculated as $E = 2.303\ k \times$ slope.

4. **Diagram:**



Energy Band gap of Semiconductor

5. Procedure:

1. Switch on the trainer kit.
2. Insert the diode and the thermometer in close contact with each other inside the glass tube. Clamp the tube immersing it in the water in the water heater.
3. Select the reverse bias voltage V_r of 2 volts at the input of the circuit.
4. Connect the micro ammeter and the diode in the circuit.
5. Note down the room temperature and the reverse saturation current.
6. Switch on the water heater.
7. Note down the magnitude of the reverse current I_S at various temperatures keeping the reverse voltage constant. Take readings at temperature intervals of 5 ^{0}C up to about 75 ^{0}C. Plot the graph of log (I/T) -vs- (1/T), determine its slope and calculate the energy gap E_g.

6. Observations:-

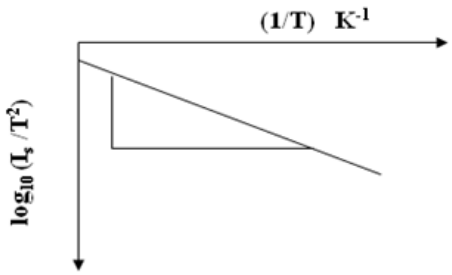
V_r= 2 volts

Obs. No.	Temp. T ^{o}C	Temp. T (K)	Rev. Sat. current I_s (µA)	$(I_s/T^2) \times 10^{-6}$ (A / K^2)	log $_{10}(I_s/T^2)$
1	75				
2					
3					
4					
5					
6					

7. Calculations and Graph:

$$E_g = -2.303 \times K \times Slope$$

$$E_g = \frac{-2.303 \times 1.38 \times 10^{-23} \times Slope}{1.6 \times 10^{-19}}\ eV$$



8. **Results:** The band energy gap of Germanium is determined to be = Eg= …… eV.

9. **Conclusion and Discussion:**
   A small rise in temperature shows a large change in reverse saturation current showing it to be highly temperature sensitive.

Faculty Signature

10. **QUIZ / Viva Questions:**
    1. Explain Diffusion and Drift Current?
    2. What is impact of reverse bias on p-n junction?
    3. What is importance of band gap in materials?

Experiment 6

1. **Aim:** To calculate the Hall coefficient and the carrier concentration of the sample material.

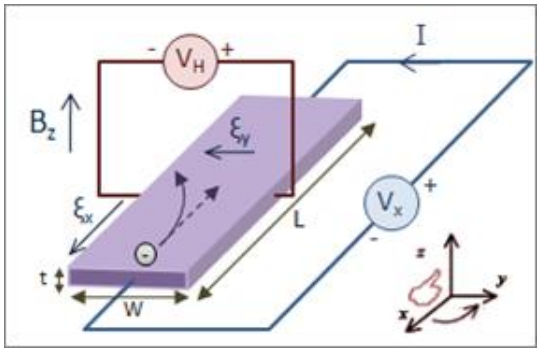
2. **What will you learn by performing this experiment?**
   Student will understand effect of magnetic field on charge carriers flowing through conductors and semiconductors**.** p type and n type semiconductor can be find out. Also carrier concentration can be calculated by using this experiment. Working principle of magnetic field sensor will also understand.

3. **Apparatus:** Electromagnet, Constant current supply, four probes, Digital gauss meter, Hall Effect apparatus (which consist of Constant Current source, digital milli-voltmeter and Hall probe).

4. **Theory:**
   If a current carrying conductor placed in a perpendicular magnetic field, a potential difference will generate in the conductor which is perpendicular to both magnetic field and current. This phenomenon is called Hall Effect. In solid state physics, Hall Effect is an important tool to characterize the materials especially semiconductors. It directly determines both the type and density of charge carriers in a given sample.

   Consider a rectangular conductor of thickness t kept in XY plane. An electric field is applied in X-direction using Constant Current source, so that current I flow through the sample. If w is the width of the sample and t is the thickness. Therefore current density is given by

$$J_x = I/wt$$



Hall Effect

If the magnetic field is applied along positive z-axis, the Lorentz force given by

$$f_m = q(v \times B)$$

deflects the charge carriers (i.e. electrons) towards the negative y-direction causing an accumulation of charge carriers the left edge of the sample. This sets up a transverse electric field E_y in the sample developing a potential difference along the y-axis. This is known as Hall voltage V_H and this effect is called Hall Effect.
In steady state condition, the magnetic force is balanced by the electric force.
Mathematically we can express it as

$$eE_y = evB_z$$

Where e is the electric charge, Ey is the Hall electric field developed, B_z is the applied magnetic field and v is the drift velocity of charge carriers. And the drift current I_x can be expressed as,

$$I_x = nevA$$

Where n is the number density of electrons in the conductor of length l, width w and thickness t.
Using above equations the Hall voltage V_H can be written as,

$$V_H = wE_y = vwB_z = \frac{I_x B_z}{net}$$

$$V_H = R_H \frac{I_x B_z}{t}$$

Where, R_H is Hall Coefficient,

$$R_H = 1/ne$$

Formula:

$$\boldsymbol{R_H = \frac{V_H t}{I_x B_z}} \quad \textbf{(}\Omega\ \textbf{m}^3\textbf{/ Wb) or (m}^3\textbf{/C)}$$

$$\boldsymbol{n = \frac{1}{eR_H}} \quad \boldsymbol{/m^3}$$

5. Procedure:

1. Adjust the required gap between the electromagnets.
2. Connect the Hall Probe in to the probe socket.
3. Adjust probe current to zero and magnetic current to zero and observe V_H=0 volt.
4. Keep the magnetic current constant. Vary the probe current from -20 mA to +20 mA in steps of 2 mA. Note down the corresponding Hall voltages.
5. Then calculate Hall coefficient and carrier concentration using the equations.
6. Plot graph of Magnetic Field V_H -vs-I and calculate Hall Coefficient.

6. Observations:

B = ……gauss =**…….** wb/m^2 **Thicknss t =** …….. mm

Sr. Number	Probe Current I (mA)	Hall Voltage V_H (mV)	Hall Coefficient R_H (ohm-m^3/wb) x 10^-5	Charge Density n (/m^3) X 10^24
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

7. Results: Hall Coefficient of given material is R_H =..... **(Ω m^3/ Wb) or (m^3/C)**

8. Conclusion and Discussion:

1. Specimen used in this experiment is..... type

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9. QUIZ / Viva Questions:

1. Define Hall Effect, Fleming's Left Hand Rule, Hall co-efficient.
2. Why is Hall potential developed?
3. How will you indentify n type or p type semiconductor?

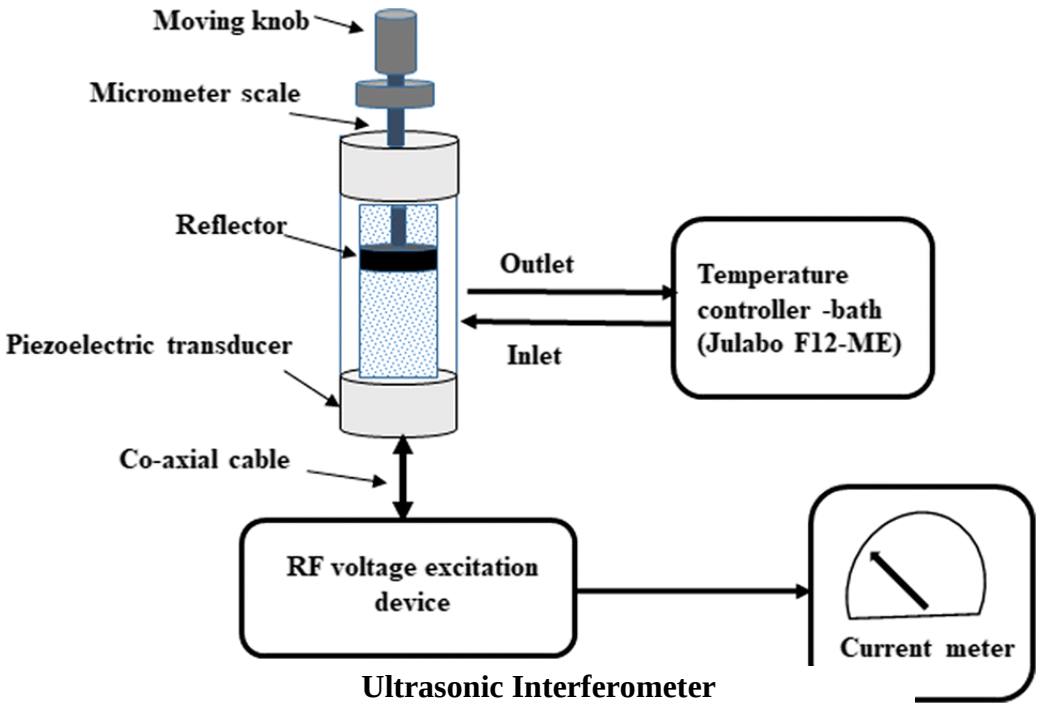
Experiment 7

1. **Aim:** To calculate the velocity of ultrasonic waves in water using ultrasonic interferometer

2. **What will you learn by performing this experiment?**
   The velocity of propagation of ultrasonic waves depends on the medium and frequency. Its applications in metallurgy, marine technology and medical field are based on this principle.

3. **Apparatus:** Ultrasonic Interferometer

4. **Theory:**
   The schematic diagram of an ultrasonic interferometer is shown in the figure.



Ultrasonic Interferometer

In an ultrasonic interferometer, the ultrasonic waves are produced by the piezoelectric method. In a fixed frequency variable path interferometer, the wavelength of the sound in an experimental liquid medium is measured, and from this one can calculate its velocity through that medium. The apparatus consists of an ultrasonic cell, which is a double walled brass cell with chromium plated surfaces having a capacity of 10ml. The

double wall allows water circulation around the experimental medium to maintain it at a known constant temperature.

The micrometer scale is marked in units of 0.01mm and has an overall length of 25mm. Ultrasonic waves of known frequency are produced by a quartz crystal which is fixed at the bottom of the cell. There is a movable metallic plate parallel to the quartz plate, which reflects the waves. The waves interfere with their reflections, and if the separation between the plates is exactly an integer multiple of half-wavelengths of sound, standing waves are produced in the liquid medium. Under these circumstances, acoustic resonance occurs. The resonant waves are a maximum in amplitude, causing a corresponding maximum in the anode current of the piezoelectric generator. If we increase or decrease the distance by exactly one half of the wavelength ($\lambda/2$) or an integer multiple of one half wavelength, the anode current again becomes maximum. If d is the separation between successive adjacent maxima of anode current, then,

$$d = \frac{\lambda}{2}$$

The velocity ($\boldsymbol{v}$) of a wave is related to its wavelength ($\boldsymbol{\lambda}$) by the relation,

$$v = \lambda f$$

where $\boldsymbol{f}$ is the frequency of the wave.

Then,

$$v = \lambda f = 2df$$

where v is the velocity of the sound wave through that medium.

5. Procedure:

1. Select the desired frequency of the generator.
2. Move the micrometer screw slowly in either clockwise or anticlockwise direction till the anode current shows an extreme deflection.
3. Note down the micrometer reading for the consecutive maximum (or minimum) current. Take 20 readings.
4. The difference of consecutive readings is a measure of λ /2.

6. Observations:-

Frequency, f = MHz

Sr. No.	Micrometer reading	λ /2 (mm)	λ (mm)	Velocity v = fλ (m/sec)	Average velocity ,v (m/sec)
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					
9.					
10.					
11.					.....
12.					
13.					
14.					
15.					
16.					
17.					
18.					
19.					
20.					

7. Results: The velocity of ultrasonic waves in water = m/sec

8. Conclusion and Discussion:

The velocity of ultrasonic waves / sound waves increases with increasing density of medium.

Faculty Signature

9. QUIZ / Viva Questions:

1. What are ultrasonic waves?
2. What are the properties of ultrasonic waves?
3. How ultrasonic waves are produced?

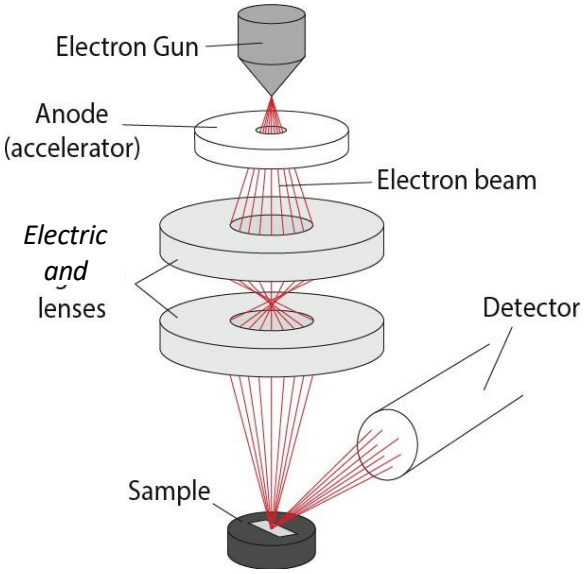
Experiment 8

1. **Aim:** Study of working principles of Scanning Electron Microscope (SEM) and Atomic Force Microscopy (AFM).

CO6	Understanding of working principles of characterizing tools used for NEMS (Nano Electro Mechanical System).

2. **Theory:**

Scanning Electron Microscopy:



Optical Microscopes give resolution of object up to micrometer range (~1 µm) as wavelength of visible light is much larger (~ 0.5 µm). In order to observe objects smaller than micrometer e.g. nanometer, we have to use source of smaller wavelength. For such small wavelength beam of electron can be used due to its wave nature given by de Broglie hypothesis,

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m_e eV}} = \frac{12.27}{\sqrt{V}} \text{ Å}$$

Scanning Electron Microscope (SEM) consist of Electron gun, Focusing lenses (condensing lenses), Scan coils, Objective lens, sample holder and electron detectors.

Principle:

A beam of electrons is incident on nanoparticles, scattered electrons from nanoparticles such as secondary electrons, backscattered electrons and x rays are detected with respective detectors. Morphology of nanoparticles is formed on monitor by secondary and backscattered electrons. Chemical composition or atomic composition is analysed by x ray detector.

Working:

A beam of electrons produced from electron gun is focused by two focusing lenses (electrostatic and magnetic). Focused beam of electrons is moved (scanned) over nanoparticles using scanning coils (deflection coils). Scanning of beam over sample holder is used to observe complete view of the nanoparticles and it also observes different regions of the nanoparticles. Finally beam is focused on sample (nanoparticles) using objective lens. Objective lens magnify image of the nanoparticles. When beam of electron incident on nanoparticles, it gets scattered. Scattered electrons consist of backscattered electrons (i.e. electron get scattered elastically with same energy by repulsion from inner shell electrons). Scattered electrons also give secondary electrons (i.e. electrons from the atoms of the nanoparticles which inelastic scattering having less energy than incident beam). Detection of backscattered electrons and secondary electrons give image formation of the nanoparticles on the monitor.

High energy electron beam produces characteristic x rays from atoms of the nanoparticles. These x rays provide chemical or atomic composition of nanoparticles. To observe nanoparticles under SEM, nanoparticles must be conducting. Non conducting samples are coated with layer of Platinum or Gold.

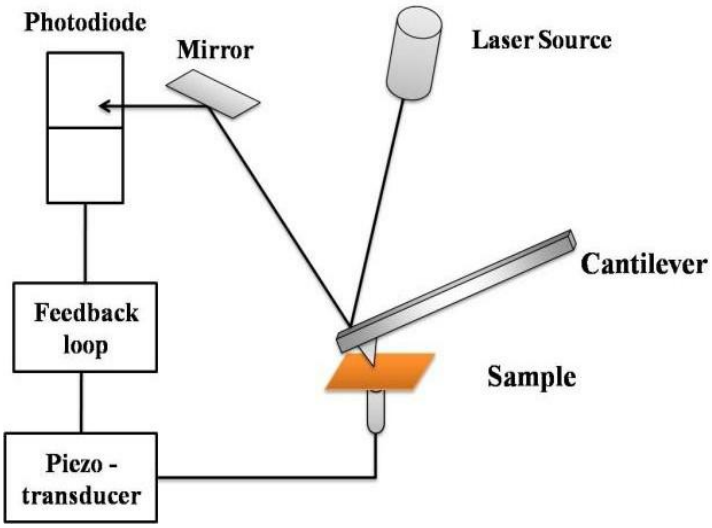
SEM gives image resolution up to 50 nm. Field Emission Scanning Electron Microscopy (FESEM) allows ~1 nm image resolution.

Atomic Force Microscopy (AFM)

Atomic force microscopy is introduced by G. Binning and C. Gerber in 1985 to overcome non conducting nature of sample in STM. Basic principle of this technique is intermolecular forces between two atoms. When two atoms come close to each other, they will experience repulsive and attractive forces depending upon distance between them. Combined force is given by following equation,

$$F = \frac{A}{R^{12}} - \frac{B}{R^{6}}$$

First term of the equation denotes repulsive force and second term denotes attractive force, where A and B are constants and R is distance between two atoms. It can be seen from the equation that repulsive forces are effective when atoms are very close to each other. Repulsive forces are due to repulsive interaction between electron clouds in atoms.



Working:

AFM consist of atomically sharp and hard tip made from Silicon Nitride (Si3N4) or Silicon (Si) mounted on head of the cantilever. This tip is moved over surface of nanoparticles. Force experienced by tip is used to study surface morphology of nanoparticles. Feedback mechanism is provided with piezoelectric sensor to maintain tip at constant height from the sample.

Deflection of cantilever due to attractive and repulsive force is detected by laser beam focused on head of cantilever and further beam is detected by photodiode (detector). Variation in current of photodiode is feedback to piezoelectric sensor to maintain height and variation in current and it is also used to study surface morphology of nanoparticles. AFM works in two modes:

1. Contact mode: In contact mode force experienced by tip is repulsive force.
2. Non-contact mode: In non-contact mode force experienced by tip is attractive.

Different intensities of laser beam detected by photodiodes are used to study surface morphology of nanoparticles.

Advantages:

1. AFM gives ~1nm image resolution.
2. Any sample conducting or insulating can be scanned by AFM.
3. Atomic resolution is obtained by AFM.
4. AFM does not require vacuum so easier to mount sample.
5. AFM gives 3D topography of surface of nanoparticles.

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QUIZ / Viva Questions:

1. Why instead of light, electron beam is used to study nanoparticles?
2. What is advantage of AFM over Electron microscopes?
3. State applications of nanotechnology in various fields.

Evaluation Report

Experiment No:	
Title :	
Student Name:	
Roll No.:	
Batch:	
Date of Conduction:	
Date of Submission:	
Grade:	
Signature of the Faculty with date	